

Deal & LP Origination Tool

Complete Product Flow Map & Business Rules

Version 2.0 | 8 July 2026 | All rules verified against production code | Supersedes v1.0 (6 July) | CONFIDENTIAL

What the product does

Two AI-powered origination engines in one tool. **Deal side:** sources, qualifies, enriches, scores and manages B2B SaaS investment targets — UK/Ireland, founder-led or bootstrapped preferred, revenue sweet spot **£2.5-10M** (hard cap £50M), EBITDA £1-10M, fit valuations £15-50M. **LP side:** finds and manages potential investors — family offices, funds of funds, HNWI's/UHNWI's — to invest through Averroes.

THE ONE ARCHITECTURE RULE THAT GOVERNS EVERYTHING

Ingest saves RAW data only — no AI at ingest. SmartFill (companies) and InvestorFill (LPs) are the ONLY AI layers, triggered per-record on demand, or in bulk behind a cost-estimate gate.

The two loops



Three screens

Screen	Purpose
Master Universe	Every company ingested. Search/filter, Sources overlay, Bulk SmartFill with cost gate, outreach. Hover any column header for its definition.
Deal Pipeline	Kanban of active deals. Qualified column sorted by fit score; later columns by latest stage entry. Cards flag Δ after 10 days in a stage.
Investors (LPs)	LP universe: fit scores, PitchBook data, PSC principals, relationship stages, bulk research, CSV shortlist export.

What changed since v1.0

Revenue sweet spot recalibrated to £2.5-10M with a 4-rule estimation engine (§5) · Revenue Band column · stringent Companies House matching (§4) · CH registry intelligence: PSC ownership proof, secured debt, quiet-raise detection, accounts-due (§4) · stage timestamps + 10-day stale flags (§7) · outreach guardrails (§7) · entire Investor/LP module (§8) · all fabricated demo data removed — every source is now real or honestly empty · standalone "Requalify All" removed (lives inside Criteria Chat apply only).

1 · End-to-End Flow — Deal Side

From raw source to signed deal. All sources are REAL scrapers or uploads; demo/fallback data has been removed product-wide.

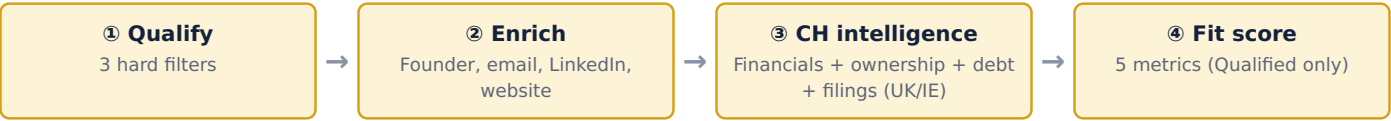
Stage A — Getting companies in (NO AI)

Source	What it actually does	Status on save
Excel/CSV upload	"pitchbook" in filename → PitchBook parser (45+ fields); else generic parser	Uploaded
SaaStock Europe	Official machine-readable archives, editions 2022-2025: sponsors + founder speakers (contacts captured)	Scraped
London Tech Week 2026	Full exhibitor list (~250) + speaker companies from the official site	Scraped
EF Alumni	Entrepreneur First portfolio directory; London companies by default; founder + LinkedIn captured	Scraped
Tech Nation Future Fifty	Cohort lists 2025 + 2026 with websites and descriptions	Scraped
Startups 100 UK	Live scrape of startups.co.uk annual list (latest year)	Scraped
TheSaaSDirectory	Page-by-page directory scrape	Scraped
Marketplaces (Acquire, Flippa, Microns, SideProjectors) · FT 1000 · Deloitte Fast 50	Not scrapeable (auth walls / paywalls / JS) — return empty honestly, never demo data	—

Stage B — The universe (holding pen)

Everything sits raw in BigQuery **targets** until SmartFill. Dedup merges re-scrapes into existing records. The first-added timestamp (**ingested_at**) is written once and never changes.

Stage C — SmartFill (per company, or in bulk)



Bulk SmartFill (Universe header): shows a funnel first — total → minus non-UK/IE → minus non-tech → minus already processed → eligible — plus a Gemini cost estimate (3-7 calls each, ~\$0.015 tokens; grounding free ≤ ~500 companies/day). Runs sequentially with progress bar and stop button. Filters are keyword-based (zero AI): companies with no region/sector text are excluded — SmartFill them individually to override.

Stage D — Working the deal



Every stage move stamps two timestamps (\$7) and logs to activity_log. Cards turn ⚠ **red after 10 days** in a stage.

Where data lives

Store	Holds
BQ · targets	Company universe, 90+ columns (PitchBook, CH financials + registry intel, scores, bands, stage timeline)
BQ · investors	LP universe, 40+ columns (PitchBook LP data, fit scores, PSC principals, relationship stages)
BQ · activity_log / qualification_config	Audit trail of stage moves & notes · versioned qualification criteria
GCS bucket	Uploaded files, scraped JSON, CH filing PDFs (ch-filings/..., viewable from the company card)

2 · Inside SmartFill — The Decision Flow

What happens when you press the button, including every branch condition.

1

Qualification — 3 hard filters

IF Gemini key set → AI qualification, ELSE keyword fallback

Geography (UK/Ireland), Industry (tech/SaaS keywords), Size (revenue < £50M; asset-aware AI estimate when undisclosed). All three pass → **Qualified**; any failure → **Not a Fit**. Full rules on page 4.

2

Enrichment — always runs

RUNS EVEN IF NOT A FIT

Gemini + Google Search: founder/CEO, contact email, LinkedIn, website, description. Saved regardless of outcome.

3

Companies House intelligence

ONLY IF geography passed (UK/IE) — even if Not a Fit on industry/size

Financials: stringent match (page 5) → up to 3 filings parsed → 3 years of Revenue, **Gross Profit** (new), Profit, Assets, Net Assets, Cash, Employees; first PDF stored in GCS. **Registry intel (new):** PSC ownership proof, outstanding charges + lenders, last share allotment (quiet-raise detector), next accounts due date.

4

Averroes Fit Scoring

ONLY IF status = Qualified

5 metrics, 0-1 each; composite = average of available. Hard rule: **4 of 5** assessable or the score is **null**. Revenue comes from the 4-rule estimation engine (page 6). Full model page 7.

5

Write-back — single atomic UPDATE

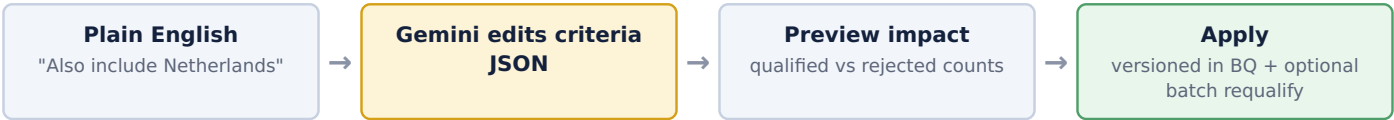
Status, size bucket, revenue band + estimate + provenance, contacts, all CH fields, all scores, stage timestamps.

DESCRIPTION RULE a new description only replaces a longer existing one — richer data never overwritten by thinner.

Outcome matrix

Case	Status	Contacts	CH intel	Fit score	Revenue band
UK/IE + tech + right size	Qualified	✓	✓	✓ (if ≥4 metrics)	✓
UK/IE, wrong industry or too large	Not a Fit	✓	✓ (still fetched)	—	✓ (local rules only)
Right industry, outside UK/IE	Not a Fit	✓	— (skipped)	—	✓ if data allows
Qualified but <4 metrics assessable	Qualified	✓	✓/—	null — unscored	✓

Changing the rules — Criteria Chat



Changed in v2: the standalone "Requalify All" button was removed — it reset pipeline-stage progress and duplicated Criteria Chat's apply step, which is now the only batch requalification path (keyword-based, stamps stage timestamps on any status change).

3 · Qualification — The 3 Hard Filters

All three must pass. Criteria live in BigQuery (qualification_config), editable in plain English via Criteria Chat; defaults below.

Filter 1 — Geography: UK or Ireland?

Fields checked

regionhq_countryhq_cityhq_location

Match list: UK, United Kingdom, England, Scotland, Wales, Northern Ireland, Ireland, London, Manchester, Birmingham, Leeds, Bristol, Edinburgh, Glasgow, Dublin, Belfast, Cambridge + codes GB, UK, IE, IRL. With Gemini, HQ can be inferred from context when fields are empty.

Filter 2 — Industry: technology / SaaS / software?

Fields checked

sectordescriptionkeywordsverticalsindustry_groupemerging_spaces

33 tech keywords: software, saas, platform, cloud, tech, digital, data, analytics, ai, machine learning, automation, cyber, fintech, healthtech, edtech, proptech, insurtech, regtech, legaltech, martech, adtech, devops, api, microservices, ml, deep learning, nlp, computer vision, robotics, iot, blockchain, web3, ecommerce. Traditional industries without tech as core product do NOT count.

Filter 3 — Size: revenue under £50M? (qualification cap — distinct from the £2.5-10M band)

Bucket	Revenue	Qualifies?	Typical profile
Micro	< £5M	YES	<30 employees, early stage
Small	£5M – £15M	YES	30–100 employees, bootstrapped-profitable or Series A/B
Mid	£15M – £50M	YES	100–500 employees, established product
Large	> £50M	NO	Too big for the thesis

Size source cascade: PitchBook revenue → CH filed revenue → Gemini estimate using total assets as primary proxy (sector-aware), employees, funding, valuation. Conservative bias: mixed signals lean smaller; undecidable defaults to "Small" (benefit of the doubt). Keyword-fallback mode with unknowable size: geography + industry decide alone.

Revenue Band — the triage layer on top (informational, does not affect qualification)

Band	Revenue (actual or estimated)	Meaning
Too Early	< £2.5M	Below the sweet spot — watchlist
Target Band	£2.5M – £10M	The sweet spot (≈ valuations £15–50M at ~6× revenue)
Too Large	> £10M	Above sweet spot; still Qualified up to the £50M cap

Shown as a colour chip in the Universe table and on Kanban cards. Estimated revenue displays as "≈£XM (est.)" with full provenance on hover.

4 · Companies House — Matching & Registry Intelligence

UK/Ireland companies only. Rebuilt in v2: a wrong company's accounts are worse than none.

Stringent matching — name similarity is a HARD GATE (v2)

Generic words (technology, software, app, solutions, ltd...) carry **zero** matching weight. A candidate must pass one of these gates or it is discarded — status/SIC/address bonuses can never rescue a wrong name:

Gate	Example	Base score	Confidence
Exact name	"Zayndu Ltd" = "ZAYNDU LTD"	100	HIGH
Same distinctive words (suffixes ignored)	"Zayndu" = "ZAYNDU LTD"	90	HIGH
One distinctive name contains the other	"Monzo" c "MONZO BANK LIMITED"	75	MED
Near-identical spelling (typos)	"Chatermill" ≈ "CHATTERMILL LTD"	65	MED
Shared distinctive word + strong similarity	partial overlap	50	LOW

Tie-breakers among gate-passers only: Active +10 · Dissolved −15 · SIC/tech alignment +5 · UK address +5. No candidate passes → **no CH data at all**. Hopelessly generic names ("Technology Group Ltd") refuse to match by design.

Financial extraction (per matched company)

Up to 3 accounts filings downloaded; Gemini parses the real PDFs → 3 years of **Revenue · Gross Profit (new in v2) · Profit · Total Assets · Net Assets · Cash · Employees** + filing type. First PDF stored in GCS, viewable from the company card ("View CH Filing PDF"; companies without a stored PDF link to their public CH filing history instead). Gross profit matters: most small UK accounts hide turnover but disclose GP — the revenue engine's Rule 2 (page 6).

Registry intelligence (new in v2 — all free API calls)

Signal	What it tells you	Example output
PSC register	Who legally owns the company — verifies "founder-led / bootstrapped" from the register, not from guesswork; detects VC entities on the cap table	"Founder/family-owned (75–100% individual control)" · "Institutional investor on register: Balderton Capital II LP"
Charges register	Outstanding secured debt and who holds it	"1 outstanding — held by HSBC UK Bank Plc"
Share allotments (SH01)	Quiet-raise detector — they issued shares with no press	"Last share issue: 2025-11-03"
Accounts next due	When fresh financials will land — re-SmartFill then	"Next accounts due: 2026-09-30"

All shown in the company drawer's Companies House section.

Consequence of a failed match: no CH data → no filed revenue or gross profit → the Revenue Growth metric is unavailable and the revenue estimate falls to proxies (page 6). The company must reach 4/5 via other metrics or its fit score stays null.

5 · Revenue Estimation Engine (v2)

How revenue is determined when it isn't disclosed. Four rules, applied strictly in order — the cascade never mixes tiers upward.

- 1

Filed revenue wins outright HIGH
CH filed turnover first, then PitchBook revenue. Real numbers are never blended with estimates.
- 2

Filed gross profit overrides all inference MEDIUM
Small accounts often hide turnover but show gross profit — filed data plus exactly one assumption: **revenue = GP ÷ sector gross margin** (software 80% · services 50% · unknown 70%). The proxy jury still runs as a sanity check; disagreement beyond 2× is noted in the provenance, but filed data keeps priority.
- 3

Median of all available proxies — a jury, not a single witness MED if 2+ agree within 2× · else LOW
Each proxy errs in a different direction per company type; the median discards outlier voices.
- 4

AI web search joins weak juries MED if it corroborates · LOW alone
Gemini + Google Search joins the median only when local evidence is weak (single proxy, or >2× disagreement, or nothing at all). Never overrides filed data. Bulk pre-filter paths stay AI-free.

The proxy jury (Rule 3)

Proxy	Conversion
Employees (PB/CH)	× £100K/head
Total assets (CH)	× sector multiplier
Est. EBITDA	× 6.5 (≈15% margin)
Valuation	÷ 6
Last funding round	× 3

Asset multiplier by sector (SIC-aware)

Sector	×
Software/SaaS (SIC 62xx, 631x)	3.0
Services/consulting	2.5
Unknown	2.0
Property/holding/finance (64xx, 68xx)	skipped

Rationale in one line each

Employees: UK B2B software runs £80–150K revenue/head; £100K is the cautious end.
Assets: asset-light software turns assets 2–4× — receivables anchor assets to billing.
EBITDA ×6.5: inverts ~15% margin, mid-band for lower-mid-market software.
Valuation ÷6: private B2B software prices at 4–8× revenue; 6 matches our £15–50M ↔ £2.5–10M view.
Round ×3: growth rounds ≈ 0.5–1× forward revenue; crudest voice, muted by the median.

Cash rule: if cash > 60% of total assets (the "just raised" signature), the asset proxy uses **assets – cash** — a funded startup's £10M cash pile no longer masquerades as £25M of revenue.

Worked examples

Filleted accounts, SIC 62012: GP £2.1M ÷ 0.80 → **£2.6M, MEDIUM, Target Band.**
Micro-entity, 3 signals: 40 staff (£4.0M) · assets £1.5M×3 (£4.5M) · val £30M÷6 (£5.0M) → median **£4.5M, MEDIUM.**
Post-raise startup: assets £10M, cash £8M → cash rule → £2M×2; web estimate £3.5M joins → **£3.8M, MEDIUM.**

THREE PRINCIPLES BEHIND EVERY NUMBER
① **Filed beats derived, derived beats inferred, inferred beats searched.** ② **Estimates are never silent: "£XM (est.)" with full recipe and confidence, everywhere.** ③ **Insufficient evidence returns LOW or blank — never a confident-looking guess.**

6 · Averroes Fit Score — The 5-Metric Model

Qualified companies only. Composite = average of available metrics; minimum 4 of 5 assessable or the score is null.

#	Metric	Source	How it's judged
1	Revenue Growth	Local — CH filings	YoY filed revenue. <−20%: 0.0 · −20-0%: 0.1-0.2 · 0-10%: 0.2-0.5 · 10-25%: 0.5-0.75 · 25-50%: 0.75-0.9 · 50%+: 0.9-1.0
2	Revenue Size	Revenue engine (p.6)	Fit to the £2.5-10M sweet spot — bands below
3	Employee Growth	Gemini + web search	LinkedIn, job postings, hiring news. Shrinking 0-0.2 · flat 0.2-0.4 · stable 0.4-0.6 · healthy 0.6-0.8 · rapid 0.8-1.0
4	Business Model Fit	Gemini + web search	B2B? SaaS/recurring? Tech core? Pure B2B SaaS 0.8-1.0 · tech-enabled services 0.4-0.6 · B2C/non-tech 0-0.2
5	Market Sentiment	Gemini + web search	Press, awards, reviews, brand. Strong 0.8-1.0 · neutral 0.4-0.6 · negative 0-0.2

Revenue Size — v2 curve (flat top across the whole sweet spot, continuous slopes)

Revenue	Score	Read
< £1M	0.10 - 0.30	Too early
£1 - 2.5M	0.30 → 1.00 rising	Approaching
£2.5 - 10M	1.00 flat	Sweet spot — all equal
£10 - 20M	1.00 → 0.50 declining	Above target
£20 - 50M	0.50 → 0.30 declining	Large end of the cap
> £50M	0.20	Shouldn't normally be Qualified

Integrity rules

① Minimum 4 of 5 metrics assessable, else score = null ("insufficient data" — never a guess). ② Missing metrics are excluded, never counted as zero. ③ All metrics must cite evidence found via live search. ④ Every score carries a written explanation, visible in the company drawer.

Reading the composite on screen

Fit score	Badge	Interpretation
≥ 0.70	GREEN	Strong thesis fit — prioritise
0.40 - 0.69	AMBER	Partial fit — review the per-metric breakdown
< 0.40	RED	Weak fit despite passing hard filters
null	UNSCORED	<4 metrics assessable — needs more data, not a judgment

7 · Deal Lifecycle, Timestamps & Outreach

The 10 statuses, the stage-timeline rules (new in v2), and the guard-railed outreach engine.

Stage timeline rules (v2)

Rule	Behaviour
First-entered timestamp	ingested_at written once at first insert — survives re-scrapes, merges and enrichment, on both companies and LPs
Current-stage timestamp	stage_entered_at resets on every stage change — drives sorting and the stale flag
Permanent per-stage stamps	qualified_at, contacted_at, meeting_at, dd_at, offer_at, won_at, lost_at — stamped on FIRST entry, never overwritten. The Contacted date survives all later moves
Every path stamps	Kanban drag, SmartFill, Criteria-Chat batch requalify, outreach-send (also stamps contacted_at), pipeline removal
Kanban sorting	Qualified column: fit score descending. Every other column: latest stage entry first
Stale flag	Card badge shows days-in-stage; > 10 days → red ⚠ badge + red card border

The 10 statuses

Status	Set by	In Kanban?
Scraped / Uploaded	Ingest (automatic)	No — Universe only
Not a Fit	SmartFill fail · Criteria apply · manual Remove (also zeroes score; never deletes)	No
Qualified → Contacted → Meeting → DD → Offer	SmartFill pass, then manual drag (all logged)	Yes
Won / Lost	Manual — closes the deal	No — archived
Engaged	Automatic only, when an outreach email sends	Yes (quirk: not manually settable)

Founder outreach — the guard-railed engine (rewritten in v2)



Guardrail	Rule
Voice	Experienced PE partner, plain English, short sentences. Intent stated honestly: "typically taking a significant stake" — neither vague "partnership" nor acquisition-first language
Truth	Never invent facts or news; never quote the founder's own financials back at them (reads as surveillance)
Banned	"I hope this finds you well", "I came across", "I was impressed", "exciting journey", "touch base", "synergies", "leverage", "utilise", "ecosystem", "best-in-class", "streamline", "scalable", "robust", "unlock"... no exclamation marks, no em dashes anywhere , no triple-adjective lists
Thin data	Record with <2 real signals → shorter, plainer 3-4 sentence cold intro; faked familiarity forbidden
Variety	Anti-template instruction + temperature 1.0 — two founders comparing notes must not see the same skeleton

Still pending: Gmail App Password (OUTREACH_SMTP_PASSWORD) — drafting works; sends error until it's configured. Also still open: no authentication on the dashboard; CORS wide open.

8 · Investor (LP) Module — New in v2

A second origination engine: family offices, funds of funds, HNWIs/UHNWIs to invest through Averroes. Same architecture: raw sources, AI on demand.

Sources (all in the Sources overlay — the single entry point)

Source	What it does	AI cost
Mine from high-fit companies	Extracts investors from active/former-investor fields of companies scoring 40+ or Qualified; portfolio overlap = warm-intro path	Zero
PitchBook LP export upload	Real 152-column "All Columns" format; filename becomes the source label; existing records merge-fill (PB fills gaps, never overwrites); USD figures. Recommended filters printed on the card	Zero
Praxis Rock directories	Public family-office/SWF directory pages (London 146 firms, largest global, MFOs, SWFs)	Zero
Companies House registry	6 investor name patterns + SIC-code search (64303 venture/dev capital, 66300 fund management) — catches family offices whose names don't say what they are; registration numbers included	Zero

InvestorFill — research + 4-criteria LP fit (1 grounded Gemini call)

Criterion	Scoring
Geography	UK 1.0 · Ireland/W. Europe 0.8 · Saudi/GCC 0.9 · rest of Europe 0.6 · US/other 0.3
PE appetite	Proven LP commitments / direct private deals 0.9–1.0 · some private markets 0.5–0.8 · public only 0.1–0.3
Ticket fit	£250K–5M typical commitments 0.9–1.0 · £5–20M 0.6 · >£50M institutional minimums 0.2
Tech affinity	Significant B2B software exposure 0.8–1.0 · some 0.5–0.7 · none evident 0.2

Integrity: minimum **3 of 4** criteria evidenced via search or the fit score is null. Auto-advances Identified → Researched. **UK entities additionally get CH enrichment:** PSC register → the controlling individual (UHNWI discovery, fills empty Contact), active directors, and filed net assets as an AUM proxy (+1 Gemini PDF call).

Working the relationship



Feature	Rule
Table order	Chronological by first-added (ingested_at, permanent) with an "Added" column
Bulk InvestorFill	Funnel first: excludes only explicit negatives (mandate outside UK/EU/ME, stated strategies with none relevant), skips researched; unknowns pass. Cost ≈ \$0.006/investor + grounding
LP outreach	IR-tone intro email (strategy prefs / mandate / portfolio overlap referenced), editable, Gmail send → stage bumps to Contacted
Export	One click → CSV of the current filtered view, 23 columns, for partner meetings
Key PitchBook signals shown	PE strategy preferences ("None relevant" shown red — a real negative) · condensed geo mandate · Open to First-Time Funds badge · # PE commitments + total \$ committed · AUM (\$M)

9 · Scenario Playouts — How the Rules Combine

Realistic cases traced end-to-end through the v2 rules.

S1 · PitchBook upload: London B2B SaaS, £8M filed revenue

QUALIFIED + SCORED

SmartFill: geo ✓ · saas ✓ · size Small ✓ → **Qualified**. CH exact-core match (HIGH) → 3yrs accounts + PSC: "Founder/family-owned (75-100%)" — bootstrapped verified from the register. Revenue £8M = Rule 1, HIGH → revenue-size score 1.0 (flat sweet spot). Target Band chip. Likely 5/5 metrics → green badge, tops the Qualified column.

S2 · Scraped UK company, filleted accounts (turnover hidden, GP £2.1M shown)

QUALIFIED, MEDIUM-CONFIDENCE REVENUE

SmartFill: qualifies; CH match; PDF parse now extracts **gross profit** → Rule 2: £2.1M ÷ 80% = **~£2.6M (est.)**, **MEDIUM**, **Target Band**, provenance on hover. Revenue Growth unavailable (no filed turnover) → needs the 3 web metrics + revenue size = 4/5 → scored.

S3 · Post-fundraise startup: assets £10M of which £8M cash

CASH RULE APPLIED

Revenue engine: cash > 60% of assets → asset proxy uses £2M, not £10M. Single weak voice → Rule 4: web estimate joins the median. Old engine would have said ~£25M (Too Large, wrongly); v2 lands ~£4M, Target Band, MEDIUM.

S4 · "Vrinsoft Technology Inc" — generic-word CH lookalike

REFUSED TO MATCH

v1 failure, v2 fix: the old matcher matched it to "ALL EAT APP NETWORK TECHNOLOGY INCORPORATED LTD" via the word "technology" + bonuses. v2: generic words carry zero weight, no gate passes → **no CH data at all**. No data beats wrong data.

S5 · Deal sits in Meeting for 12 days

STALE FLAG

Kanban: card shows **12d** in red with a red border. Its meeting_at stamp is permanent; when it moves to DD, stage_entered_at resets but the Meeting entry date survives forever.

S6 · LP: "XYZ Investments Ltd" from the CH registry scrape

UHNWI DISCOVERED

InvestorFill: web research + CH enrichment: PSC says "controlled by John Smith (individual, 75-100%)" → Contact fills with the principal; filed net assets £45M becomes the AUM proxy. Criteria: UK 1.0 · PE appetite evidenced 0.8 · ticket unknown (null) · tech affinity 0.6 → 3/4 assessable → fit 80, green. Stage auto-moves to Researched.

S7 · Founder outreach on a thin record

PLAIN COLD INTRO

Guardrails: record has name + sector only → 3-4 sentence plain intro, no faked familiarity, no jargon, no em dashes, intent stated as "usually taking a significant stake". Send stays blocked until the Gmail App Password is set; on success → Engaged + contacted_at.

IF YOU REMEMBER ONLY FOUR THINGS

① **No AI at ingest** — SmartFill/InvestorFill are the only AI moments, always behind a visible cost when bulk. ② **Filed beats derived beats inferred beats searched** — and estimates always show their recipe. ③ **Wrong data is worse than no data: matching refuses, scores go null, sources return empty rather than fake.** ④ **Nothing is ever deleted** — records change status, every change is stamped and logged.